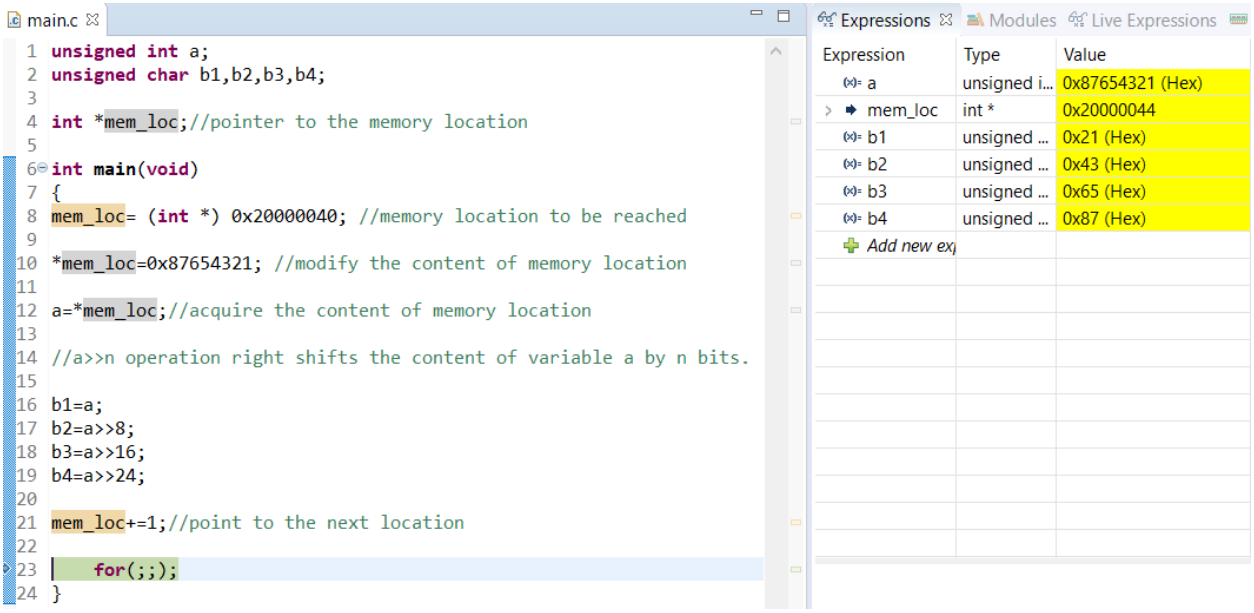
EE2004 lab session, week7, section Friday



The above code is an example on how to reach a specific memory location in C language using pointers. The only constraint here is that we can reach a memory location having content of 32 bits. To reach 8-bit values within this 32-bit content, we should apply the right shift operation.

Task:

a) Sixteen characters are saved in memory starting from 0x20000040.
b) The content will be as follows.

0xF0, 0xF1, 0xF2, 0xF3, 0xF4, 0xF5, 0xF6, 0xF7, 0xF8, 0xF9, 0xFA, 0xFB, 0xFC, 0xFD, 0xFE, 0xFF

c) Check for the memory content in character level. If a content is odd, replace it by 0x00. If the content is even, replace it by 0xFF.

Hint:

- a=a<<8; left shifts the bits in variable a by 8.
- b1 & 0x01 operation is done to apply bitwise AND operation between unsigned char variable b1 and hexadecimal number 0x01
- a|=b1 operation is done to apply bitwise OR operation between unsigned char variable a and b1. The result is saved in variable a.